Critical Defect Incident Report

Crack detected on single_crack.mp4 at 00:00.0

1. Summary

Report ID	INC-20261018-0001
Severity	**CRITICAL**
Defect type	Crack
Detection confidence	94% (best), 73% (mean)
Location	VIT Chennai, Kelambakkam-Vandalur Road, Chennai, Tamil Nadu 12.840795, 80.153556 (SIMULATED)
Recommended action	Dispatch a track patrol to verify on site. Consider a temporary speed restriction until the defect is verified.
Status	Unverified (AI-detected, pending verification)

2. Inspection Context

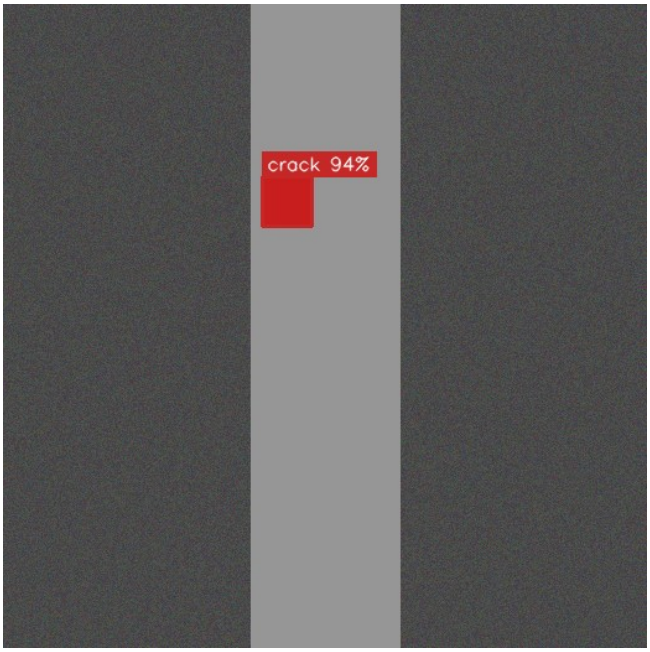
Source video	single_crack.mp4
Inspection date / time	18 Oct 2026, 17:49:27
Inspection module	Railway track defect inspection
Model	Faster R-CNN (ResNet-50 FPN) (fake_model.pth)
Weights SHA-256 (first 16)	n/a
GUI confidence threshold	0.20
Operator	root
Event detected at	18 Oct 2026, 17:49:27

3. Defect Details

Longitudinal or transverse fracture line in the rail head, typically caused by rolling contact fatigue (RCF).

Class	Crack
Bounding box (px)	x1=209, y1=139, x2=249, y2=179
Size in image	40 x 40 px (real-world size unavailable: camera not calibrated)
Frame range	0 to 139 (best frame 39)
Video time range	00:00.0 to 00:09.3
Confirmed detections	140 frames

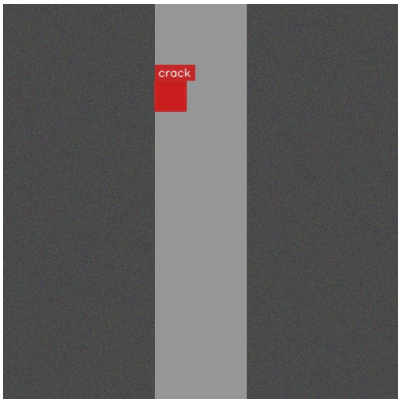
4. Evidence



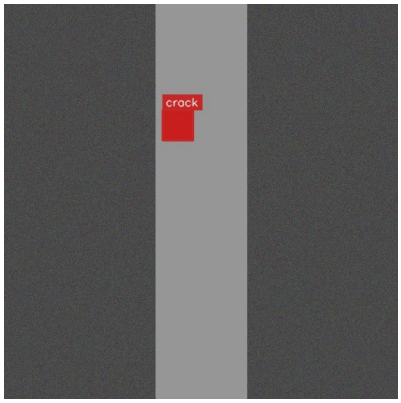
Annotated frame (best detection)



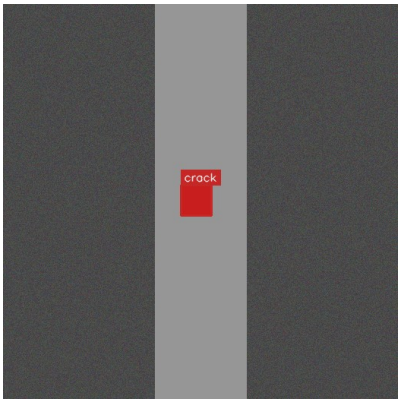
Defect crop (enlarged)



First seen (frame 0)



Best (frame 39)



Last seen (frame 139)

5. Location

Place	VIT Chennai, Kelambakkam-Vandalur Road, Chennai, Tamil Nadu
GPS (lat, lon)	12.840795, 80.153556 (SIMULATED)
Chainage / km marker	Not available
Section / line	Not available
Location source	SIMULATED: dummy telemetry, not a real position

6. Risk and Recommended Action

Risk	A propagating crack can lead to a rail break and derailment.
Recommended action	Dispatch a track patrol to verify on site. Consider a temporary speed restriction until the defect is verified.
Required response time	Per railway maintenance rules (not configured)

7. History

No previous events of this class are recorded.

Matching is by class only because the location is simulated; match by position once real GPS or chainage is available.

8. Verification and Sign-off

Reviewer name	
Role / designation	
Decision	[ ] Confirmed defect [ ] False positive [ ] Needs re-inspection
Notes	
Signature	
Date	

Appendix: Model, Limitations and Disclaimer

Model	Faster R-CNN (ResNet-50 FPN)
Model benchmark	Project benchmark on the track-fault test set: mAP50 17.69%, precision 46.61%, recall 8.63%.
Weights file	fake_model.pth
Weights SHA-256 (first 16)	n/a
GUI confidence threshold	0.20
Event confirmation rule	3 detections within 6 frames; event closes after 15 frames unseen

Limitations

- Detections come from a machine-learning model and may include false positives or miss defects.
- Confidence is the model's score, not a calibrated probability.
- Sizes are in image pixels. No real-world scale is available unless the camera is calibrated.
- Location is simulated whenever the location source says so. Do not use it to dispatch crews.
- This report must be verified by a qualified person before any maintenance or operational decision.